# Saurabh Shah

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## Education

**University of Pennsylvania** — **Philadelphia, PA**

*Master of Science in Engineering in Computer Science 3.91 GPA* — *Jan 2021 - May 2023*

- Focused on Algorithmic Theory, Artificial Intelligence, Machine Learning, and Natural Language Processing (NLP)

*Bachelor of Science in Engineering in Networked and Social Systems Engineering 3.90 GPA* — *Aug 2019 - May 2023*

- Major combines Computer Science, Systems Engineering, and Economics. Minors in Data Science and Mathematics

## Experience

**humans&** — **San Francisco, CA**

*Member of Technical Staff* — *Jan 2026 - Present*

- Built RL infrastructure, evals, and environments to train Ampersand, our first model, trained in long-horizon multi-party simulations to collaborate with groups of people

**Allen Institute for AI (Ai2)** — **Seattle, WA**

*Research Engineer (paper)* — *Feb 2025 - Jan 2026*

- Olmo team. Trained Olmo 3 to write code, use tools, and reason. Built RL + code execution infra. Code data curation

**Apple** — **Seattle, WA**

*Machine Learning Engineer* — *Oct 2023 - Feb 2025*

- Siri Natural Language Understanding (NLU). Helped build an agentic Siri planner powered by Apple Intelligence

**Allen Institute for AI (Ai2)** — **Seattle, WA**

*Research Engineering Intern (paper)* — *Aug 2023 - Oct 2023*

- Tried pretraining Olmo with ReLoRA, a parameter-efficient *pretraining* method. Learned lots about PyTorch/FSDP

**University of Pennsylvania** — **Philadelphia, PA**

*Researcher (paper) (talk)* — *Aug 2022 - May 2023*

- Explored using free-text explanations for improving the robustness of LLMs to spurious cues in training data

**Apple** — **Seattle, WA**

*Machine Learning Engineering Intern* — *May 2022 - Aug 2022*

- Siri NLU. Built an internal Swift iOS app for testing text-to-intent parsing and counterfactual evaluation

**Amazon** — **Nashville, TN**

*Software Development Engineering (SDE) Intern* — *May 2021 - Aug 2021*

- Robotics-AI Computer Vision. Built a web app with React and AWS to configure, search, and view over 300,000 cameras

## Personal Projects

**Griffin LM + CUDA (link)** — **May 2024 - August 2024**

- I learned some cuda (link) and tried to implement Griffin from scratch in PyTorch with a cuda extension for the scan

**Concept Space Embeddings (link)** — **Feb 2023 - Apr 2023**

- Worked with a team of 2 to create a novel method for interpretable embeddings of arbitrary text using LLMs and Decision Trees. Works for classification, regression, clustering, and post hoc explanation of black box models

**Compass (Penn Course Recommendation) (link)** — **Jan 2023 - Apr 2023**

- Group of 4. Course recommendation web app. I built the recommendation system with (1) collaborative filtering and (2) text embeddings to recommend courses to students based on (1) perceived difficulty and (2) natural language interests

**Poké-GANs (Pokémon Generator) (link)** — **Mar 2022 - Apr 2022**

- Generated complete Pokémon from names. Fine-tuned GPT-3 for types, stats, abilities; CLIP+VQGAN for images from generated text. Trained custom LSTM and GANs from scratch and compared results. Worked with partner.

**Comedy Bot (link)** — **July 2020**

- Experimented with ML models to recognize and rate jokes I write and perform for crowds of 150+. Joke datasets from Kaggle. Experimented with Bag of Words/Naïve Bayes and LSTM models. Built with PyTorch